

BMW Group Standard

Released

GS 93008-2
2022-10

Replacement for
GS 93008-2:2021-05

Substances of concern

Process materials and auxiliary materials

Prohibited and declarable substances

Edited by

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TEREK system.

Amendments

The following amendments have been made to GS 93008-2:2021-05:

- sections 3 and 4 added;
- Table 1 a updated (see column "Last amendments");
- removed from Table 1 a:
 - Boric acid and borates deleted, reason: as of 17.12.2022 at the latest, a specific limit value no longer applies, instead the general limit value for substances toxic to reproduction (0,3%) for borates classified as harmonized (the group entry in Table 1 b on substances marked with C, M, R of category 1A or 1B (according to 1272/2008) applies.
 - Octylphenolethoxylate (Group name: "Alkyphenolethoxylate without Nonyl"), nonylphenol-ethoxylate and several phthalate ester (benzyl butyl phthalate CAS No. 85-68-7, Di C7 to C11 alkylate phthalate ester CAS No. 68515-42-4, Di C6 to C8 branched alkylate phthalate ester CAS No. 71888-89-6, dibutyl phthalate CAS No. 84-74-2, diisobutyl phthalate CAS No. 84-69-5) have been deleted, reason: the specific sunset date of the substances for annex XIV according to 1907/2006 has expired (the group entry in Table 1 b on substances listed in annex XIV according to 1907/2006 applies);
 - Disodium tetraborat pentahydrate CAS No. 12179-04-3;
 - Diisopentyl phthalate CAS No. 605-50-5;
 - Di-(2-ethylhexyl) phthalate CAS No. 117-81-7;
 - Di-n-pentyl phthalate CAS No. 131-18-0, CAS No. 84777-06-0, CAS No. 776297-69-9;
 - Dioctyltin oxide CAS No. 870-08-6;
- Table 1 b updated (see column "Last amendments");

- in Table 1 b:
 - under group name "substances according to TRGS 900 marked with H, Z " first entry removed;
 - Per- and polyfluoroalkyl substances and products containing diisocyanate added;
- Annex A updated;
- normative references updated;
- Group Standard editorially revised.

Previous editions

BMWN(S) 11389.0 T(P)2: 1997-12, 1999-08

GS 93008-2: 2002-07, 2005-01, 2006-09, 2008-03, 2009-07, 2011-11, 2012-11, 2014-09, 2016-03, 2017-03, 2018-08, 2019-09, 2021-05

1 Scope and purpose

This Group Standard serves suppliers, manufacturers and all BMW specialist departments as an instrument for management of hazardous substances, hazard analysis and sustainability of process materials and auxiliary materials.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

ChemVerbotsV	Verordnung zur Neuregelung nationaler Vorschriften über das Inverkehrbringen und die Abgabe von Chemikalien (Artikel 1 Verordnung über Verbote und Beschränkungen des Inverkehrbringens und über die Abgabe bestimmter Stoffe, Gemische und Erzeugnisse nach dem Chemikaliengesetz (Chemikalien-Verbotsverordnung – ChemVerbotsV))
Clean Air Act (US) ³⁾	Amendments of 1990 (104 Stat. 2468, P.L. 101-549)
GB 24409 (CN) ⁶⁾	Limit of harmful substances of vehicle coatings
GefStoffV ⁶⁾	Verordnung zur Neufassung der Gefahrstoffverordnung und zur Änderung sprengstoffrechtlicher Verordnungen (Artikel 1 Verordnung zum Schutz vor Gefahrstoffen (Gefahrstoffverordnung – GefStoffV))
GS 93008-4	Substances of concern; Materials and components; Constituents and emissions
HAPS ⁴⁾	List of Hazardous Air Pollutants with Modifications
TRGS 600 ⁶⁾	Substitution
TRGS 611 ⁶⁾	Verwendungsbeschränkungen für wassermischbare bzw. wassergemischte Kühlschmierstoffe, bei deren Einsatz N-Nitrosamine auftreten können
TRGS 615 ⁶⁾	Verwendungsbeschränkungen für Korrosionsschutzmittel, bei deren Einsatz N-Nitrosamine auftreten können
TRGS 900 ⁶⁾	Workplace limit values
TRGS 905 ⁶⁾	Verzeichnis krebserzeugender, keimzellmutagener oder reproduktionstoxischer Stoffe
TRGS 906 ⁶⁾	Verzeichnis krebserzeugender Tätigkeiten oder Verfahren nach §3 Abs. 2 Nr. 3 GefStoffV
VKIS-VSI-IGM ⁷⁾	Stoffliste für Kühlschmierstoffe nach DIN 51385 für die Metallbearbeitung, 20. Auflage, Stand 18.12.2020
AwSV ⁶⁾	Verordnung über Anlagen zum Umgang mit wassergefährdenden Stoffen
98/70/EC	Directive 98/70/EC of the European Parliament and of the council of 13 October 1998 relating to the quality of petrol and diesel fuels and amending council directive 93/12/EEC
517/2014 ⁶⁾	Regulation (EU) No 517/2014 of the European Parliament and of the council of 16 April 2014 on fluorescent greenhouse gases
648/2004 ⁶⁾	Regulation (EC) No 648/2004 of the European Parliament and of the Council of 31 March 2004 on detergents
1005/2009 ⁶⁾	Regulation (EC) No 1005/2009 of the European Parliament and of the Council of 16 September 2009 on substances that lead to a reduction of the ozone layer

1272/2008 ⁶⁾	Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006. This regulation is representative for dozens of other country regulations that have also implemented the GHS (Globally Harmonized System of Classification and Labeling).
1907/2006 ⁶⁾	Regulation (EC) no 1907/2006 of the European Parliament and of the council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC
2000/53/EC ⁶⁾	Directive 2000/53/EC of the European Parliament and of the Council of 18 September 2000 on end-of life vehicles
2000/60/EG ⁶⁾	Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy
2019/1021 ⁶⁾	Regulation (EU) 2019/1021 of the European Parliament and of the Council of 20 June 2019 on persistent organic pollutants (recast)

3) refer to <http://www.epa.gov/air/caa/>

4) refer to <https://www.epa.gov/haps/initial-list-hazardous-air-pollutants-modifications>

6) refer to www.umwelt-online.de / "umwelt-online Login"

7) refer to <http://www.vsi-schmierstoffe.de/regelwerke/vkis-vsi-igm-stoffliste.html>

8) refer to <http://std.samr.gov.cn/gb/>

3 Abbreviations, terms and definitions

3.1 Abbreviations

Abbreviations are explained in the relevant places in this document.

3.2 Terms

There are no terms in this document.

3.3 Definitions

There are no definitions in this document.

4 Designation

There is no designation in this document.

5 Remarks

All relevant national and international laws and regulations shall be observed in the assessment and approval of process materials and auxiliary materials.

The supplier is obliged to register the chemical substances contained in the goods in accordance with the applicable laws for the respective market concerned, e.g. in accordance with Regulation 1907/2006 (REACH), and if necessary to have them approved or registered.

For the declaration of the constituents of a mixture, the safety data sheet in the national language shall be sent to the specialist department "Chemical Products Release".

If requested by the specialist department "Chemical Products Release", the form "Exact chemical composition for industrial hygiene purposes" (sample in annex A) shall be completed additionally by the supplier.

The evaluation of the chemical composition shall be carried out by the releasing specialist departments on the basis of the substances and substance groups mentioned in section 7.

In order to minimize endangerment to persons dealing with chemical products, the substances listed in section 7 in addition to legal controls are subject to a BMW internal prohibition and/or mandatory declaration.

The listing is based on:

- scientific treatises;
- internal BMW specifications for prevention and avoidance of health risks to employees and customers;
- responsibility toward the environment and BMW guidelines.

Process materials and auxiliary materials are e.g. chemical products, which remain on the vehicle (series material, e.g. paints), fluids in equipment and machines (e.g. lubricants), chemical consumables (e.g. cleaners), maintenance material (e.g. machine lacquers).

Prohibited substances (P) are prohibited in general or for certain uses or internally at BMW and shall neither be added to nor be set free by the products during their defined scope of use. If no limit values are listed the limit values for classification and marking in accordance with regulation 1272/2008 apply.

Any instances of impurities with these substances are declarable up to the limit of detection (state of the art).

Declarable substances (D) shall be declared if exceeding 0,1 % in the mixture of legally prescribed limit value. This information supports the BMW specialist departments in their comments in the hazard analysis.

For substances subject to substitution (S), a substitute material search is necessary. The employer shall always carry out a search for substitution when working with hazardous substances except in cases with low risk (TRGS 600). The evaluation is the responsibility of the assessing specialist departments according to BMW Group instruction "Chemical product release". During this procedure, all substances intended for a potential use shall be evaluated in the same way, taking into account technical, economical and ecological aspects. The specified data are the basis for the dialog between the supplier and the customer to ensure a safe and environmentally compatible application. It is not the objective to automatically substitute the declared S substances, but to jointly consider the advantages/chances and the potential risks taking into account the specific application. Only after thorough consideration, the material is finally selected.

Substances subject to mandatory substitution are either

- substances on the REACH candidate list (SVHC) $\geq 0,1$ % in the mixture according to Regulation 1907/2006 (these substances are banned for new developments, i.e. for products that have not yet been used in the BMW Group) or
- substances that lead to an overall classification of the product as listed in Table 2 (this means that the chemical product is subject to mandatory substitution).

6 Deviations

Every product change shall be clarified with the specialist department "Chemical Products Release" and requires new sampling inspections and new releases where applicable.

The following changes shall be declared:

- quantitative change: Change of the composition of a product leading to a new grading/classification or to a qualitative change as well as addition or removal of components;
- qualitative change: Change of the physical, chemical or toxicological parameter and/or characteristics of a product.

7 List of substances prohibited internally at BMW and of declarable substances

Table 1 a contains the prohibited and declarable substances in process materials and auxiliary materials. Table 1 b contains regulated substance groups and product groups. Table 2 contains substances and substance groups subject to mandatory substitution.

The substance classes listed in Table 1 a, Table 1 b and Table 2 only contain the most important members of these substance classes and shall not be viewed as exhaustive. If in doubt, the definition of the substance class in the pertinent legislation shall be referenced.

Substances subject to mandatory substitution from Table 2 are not additionally marked with an "S" in Table 1 a and Table 1 b.

The restrictions listed in the tables apply to the protection of employees and the environment at all locations worldwide, irrespective of the original, national legislation. If there is a lack of technical or economic alternatives at a site, this shall be notified if the product shall be used as a mandatory requirement.

The column "Inclusion in Group Standard" has not been filled retrospectively. No entry means, that inclusion of substance in Group Standard was before 2014-09.

Table 1 a: Prohibited and declarable substances in process materials and auxiliary materials

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classific ation
	Beta-Alanin-di-acetic acid	beta-ADA	6245-75-6		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
Amine (secondary)					Secondary amines which do not form carcinogenic N-nitrosamines.		2016-03		D
				TRGS 611; TRGS 615; VKIS-ISI-IGM	Precursors for carcinogenic nitrosamines. Regulated according to German TRGS 615 for corrosion inhibitors and German TRGS 611 for lubricants.	Corrosion inhibitors, cooling lubricants, foam materials, oxidation inhibitors in rubber, herbicides, pigments		2016-03: Classification	P
	e.g. Dicyclohexylamine	DCHA	101-83-7	ditto	No formation of carcinogenic nitrosamines of category 1A / 1B		2016-03		D
	N,N-Diethanolamine		111-42-2	ditto; Clean Air Act (US); VKIS-ISI-IGM	ditto	ditto		2016-03: Law, Classification	D, P
	N,N-Diethylamine		109-89-7	ditto	ditto	ditto			D
	N,N-Di-i-propylamine		108-18-9	ditto	ditto	ditto			D
	N,N-Dimethylamine		124-40-3	ditto	ditto	ditto			D
	N,N-Di-n-propylamine		142-84-7	ditto	ditto	ditto			D
	N,N-Di-n-butylamine		111-92-2	ditto	ditto	ditto			D
	N,N-Ethylphenylamine		103-69-5	ditto	ditto	ditto			D
	N,N-Methylethylamine	Ethylmethyl-amine	624-78-2	ditto	ditto	ditto			D
	N-Methyl-N-phenylamine	N-Methylaniline	100-61-8	ditto	ditto	ditto			D
	Morpholine		110-91-8	ditto	ditto	ditto		2016-03: Classification	D, P
	Piperidine		110-89-4	ditto	ditto	ditto			D
	Pyrrolidine		123-75-1	ditto	ditto	ditto			D

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Aminobiphenyl and its salts				1907/2006 Annex XVII No. 15	Shall not be placed on the market, or used, as substances or in mixtures in concentrations > 0,1 % mass percentage.			2014-09: Law, remark	P
	e.g. 4-Aminobiphenyl	Biphenyl-4-ylamine	92-67-1	1907/2006; Clean Air Act (US)	ditto			2014-09: Law	D, P
	2-Amino-1-Butanol		96-20-8	VKIS-ISI-IGM TRGS 900 1272/2008	Status of medical toxicological evaluation: H, Z substance according to TRGS 900 but not classified as toxic to reproduction.	Cooling lubricant	2014-09	2022-06: Law, remark	D
	2-(2-Aminoethoxy)-ethanol	Di glycol amin	929-06-6	VKIS-ISI-IGM	REACH Registration dossier mit DNEL: 0,67 mg/m³	Cooling lubricant	2014-09		D
	1-Aminopropan-2-ol		78-96-6	VKIS-ISI-IGM	Substance with limit value	Cooling lubricant	2014-09		D
	Amino-tris-methylene-phosphonic acid	ATMP	6419-19-8		complexing agent, biologically hardly degradable. sewage pollution due to heavy metal load.	Complexing agent			D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Arsenic and its compounds				1907/2006; Clean Air Act (US)	Declarable in all applications not otherwise prohibited.	Paint coats and melt coatings, leather and textile finishes, plane glass, metal adhesive, pyrotechnic objects, metal finishes		2014-09: Law	D
				1907/2006 Annex XVII No. 19	Prohibited in protective coatings, wood and constituents for water treatment	Paints and melt coatings, wood preservative, leather and textile finishes, water treatment, plane glass, metal adhesive, pyrotechnic objects, metal finishes, wood impregnation		2014-09: Law	P
	e.g. Arsenic		7440-38-2	1907/2006	ditto	ditto			D, P
	Arsenic acid		36465-76-6	1907/2006	ditto	ditto			D, P
	Arsenic acid and its salts		7778-39-4	1907/2006 Annex XIV	Arsenic acid is prohibited without authorization starting 22.08.2017	ditto		2018-08: Law, remark	D, P
	Diarsenic pentoxide		1303-28-2	1907/2006 Annex XIV	Prohibition without approval starting 21.05.2015	ditto		2016-03: Law, Remark, Classification	P
	Diarsenic trioxide	Arsenic trioxide	1327-53-3	1907/2006 Annex XIV	Prohibition without approval starting 21.05.2015	ditto		2016-03: Law, Remark, Classification	P
	Di-gallium arsenide phosphide		12044-20-1	1907/2006	ditto	ditto			D, P
	Gallium arsenide		1303-00-0	1907/2006	ditto	ditto			D, P
	Potassium arsenate		10124-50-2	1907/2006	ditto	ditto			D, P
	Potassium dihydrogen arsenate		7784-41-0	1907/2006	ditto	ditto			D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classific ation
Benzidine and its salts				1907/2006 Annex XVII No. 13	Shall not be placed on the market, or used, as substances or in mixtures in concentrations > 0,1 % mass percentage.			2014-09: Remark	P
	e.g. Benzidine	4'- Diaminobiph enyl	92-87-5	1907/2006; Clean Air Act (US)	ditto			2014-09: Law	D, P
	Salts of Benzidine		531-85-1	1907/2006	ditto				D, P
	Benzidine sulfate		21136-70-9	1907/2006	ditto				D, P
	3'-Dimethylbenzidine- dihydrochloride		612-82-8	1907/2006	ditto				D, P
	Benzene		71-43-2	1907/2006; Clean Air Act (US)	Compulsory declaration requirement for fuels subject to guideline 98/70/EC.	Fuels		2014-09: Law	D
			71-43-2	1907/2006 Annex XVII No. 5; GB 24409 (CN)	Prohibited as a substance and in mixtures (paint limit value 0,1 %) and toys (limit value 5 ppm)	Solvent for latex paints, raw material for other chemicals		2021-05: Law, remark	P
Benzene releasing substances					Substance can release benzene				D
	e.g. tert-Butyl peroxybenzoate		614-45-9		ditto				D
	tert-Amyl peroxybenzoate		4511-39-1		ditto				D
	α,α -Dimethylbenzyl peroxybenzoate		7074-00-2		ditto				D
	Di-benzoyl peroxide		94-36-0		ditto				D
	1,2-Benzisothiazol-3(2H)- one		2634-33-5	VKIS-ISI-IGM	Mandatory labeling as H317 starting 0,05 %; Biocidal active substance	Cooling lubricant	2014-09	2018-08: Remark	D
	Benzotriazole		95-14-7	VKIS-ISI-IGM	Non-ferrous metal compatibility	Cooling lubricant	2014-09		D
	Diphenyl		92-52-4	Clean Air Act (US)	limit for introduction into sewage			2014-09: Law	D
	N.N-Bis-(carboxy- methylene)-1- aminoethane-1,1- diphosphoric acid	BADP			complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent		2016-03: CAS number	D

Name of group	Substance	Synonym	CAS number	Law/Regulation/Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	1,3-Bis-(hydroxymethyl)-5,5-dimethyl-imid-azolidin-2,4-dion	DMDMH	6440-58-0	VKIS-ISI-IGM	Biocidal active substance	Cooling lubricant	2014-09	2016-03: Remark	D
	Bis (2-methoxyethyl) ether	Diethylene glycol dimethyl ether	111-96-6	1907/2006 Annex XIV	Prohibition without approval starting 22.08.2017	Paints, lacquers and hydraulic liquids		2018-08: Law, remark	D, P
		Diethylene glycol dimethyl ether	111-96-6	1907/2006 Annex XVII No. 30; GB 24409 (CN)	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility; Prohibited > 0,03 % in paints (total content ethylene glycol ether/ester)	Paints, lacquers and hydraulic liquids		2021-05: Law, remark	P
	Bisphenol A	4,4'-Iso-propylidene-diphenol	80-05-7	1907/2006	Prohibited in articles for babies, thermal paper; Declarable for all other applications. Candidate list	Manufacturing of resins		2021-05: Classification	D, P
Lead compounds				1907/2006; Clean Air Act (US)		Constituent of steel, aluminum alloys, batteries, soldering material		2014-09: Law	D
				1907/2006 Annex XVII No. 16/17/30; 2000/53/EG; GB 24409 (CN)	Prohibited as a substance and in mixtures/paints; Prohibited in vehicles (e.g.lead solder)			2022-06: Remark	P
	e.g. Lead		7439-92-1	1907/2006	ditto toxic to reproduction				D, P
	Lead arsenate		10102-48-4	2000/53/EG	ditto				D, P
	Lead diarsenate		3687-31-8	1907/2006	ditto candidate list			2021-05: Classification	D, P
	Lead hydrogen arsenate		7784-40-9	1907/2006	ditto candidate list			2021-05: Classification	D, P
	Lead acetate, alkaline		1335-32-6	1907/2006	ditto toxic to reproduction				D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Lead azide		13424-46-9	2000/53/EG	ditto toxic to reproduction				D, P
	Lead carbonate (water-free)		598-63-0	1907/2006 Annex XVII No. 16a	Prohibited in paints	Paints			D, P
	Lead chromate		7758-97-6	1907/2006 Annex XIV	Prohibition without approval starting 21.05.2015				P
	Lead chromate molybdate sulfate red	C.I. 77605	12656-85-8	1907/2006 Annex XIV	Prohibition without approval starting 21.05.2015				P
	Lead diacetate		301-04-2	1907/2006	ditto toxic to reproduction				D, P
	Lead hexafluorosilicate		25808-74-6	1907/2006	ditto toxic to reproduction				D, P
	Lead hydrogen carbonate	White lead	1319-46-6	1907/2006 Annex XVII No. 16b	Prohibited in paints	Paints			D, P
	Lead sulfate		7446-14-2	1907/2006 Annex XVII No. 17a	ditto prohibited in paints	Paints			D, P
	Lead sulfochromate yellow	C.I. 77603	1344-37-2	1907/2006 Annex XIV	Prohibition without approval starting 21.05.2015			2016-03: Remark	P
	Lead-2,4,6-Trinitroresorcinat	Lead styphnate	15245-44-0	1907/2006	ditto candidate list			2021-05: Classification	D, P
	Tri lead bis(orthophosphate)		7446-27-7		ditto toxic to reproduction				D, P
	2-(2-Butoxyethoxy)-ethanol	DEGBE	112-34-5	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
		DEGBE	112-34-5	1907/2006 Annex XVII No. 55	Putting into circulation in spray paints and cleaning sprays in aerosol dispensers as of 3 % prohibited	Solvent for paints		2014-09: Law	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Cadmium and its compounds				1907/2006; 2000/53/EG; Clean Air Act (US)	Compulsory declaration requirement for the exceptions as per 2000/53/EC Annex II	Batteries for electric vehicles (only spare parts for vehicles built before 2009)		2019-09: Example of use	D
				1907/2006 Annex XVII No. 23; 2000/53/EG; GB 24409 (CN)	Prohibition in paints; prohibition in vehicles; limit for introduction into sewage	Surface protection of metals, stabilizers in polymers, pigments in paints and plastics		2021-05: Law, remark	P
e.g.	Cadmium		7440-43-9	ditto	ditto	ditto			D, P
	Cadmium carbonate		513-78-0	ditto	ditto	ditto			D, P
	Cadmium chloride		10108-64-2	ditto	ditto	ditto			D, P
	Cadmium cyanide		542-83-6	ditto	ditto	ditto			D, P
	Cadmium diacetate		543-90-8	ditto	ditto	ditto			D, P
	Cadmium diformate		4464-23-7	ditto	ditto	ditto			D, P
	Cadmium distearate		2223-93-0	ditto	ditto	ditto			D, P
	Cadmium fluoride		7790-79-6	ditto	ditto	ditto			D, P
	Cadmium-hexafluoro-silicate (2-)		17010-21-8	ditto	ditto	ditto			D, P
	Cadmium iodide		7790-80-9	ditto	ditto	ditto			D, P
	Cadmium nitrate		10325-94-7	ditto	ditto	ditto			D, P
	Cadmium nitrate, tetra-hydrate (1:2:4)		10022-68-1	ditto	ditto	ditto			D, P
	Cadmium oxide		1306-19-0	ditto	ditto	ditto			D, P
	Cadmium selenide sulfide		12626-36-7	ditto	ditto	ditto			D, P
	Cadmium sulfate		10124-36-4	ditto	ditto	ditto			D, P
	Cadmium sulfate (1:1) hydrate (3:8)		7790-84-3	ditto	ditto	ditto			D, P
	Cadmium sulfide		1306-23-6	ditto	ditto	ditto			D, P
	Cadmium telluride		1306-25-8	ditto	ditto	ditto			D, P
	Cadmium-zinc-sulfide		12442-27-2	ditto	ditto	ditto			D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Chloroalkanes		SCCP/ MCCP/ LCCP		HAPS	Declarable in all applications		2017-03		D
	Chloroalkanes		61788-76-9	HAPS; 1907/2006; 2019/1021	ditto; Prohibited if short-chained chlorinated alkanes are present		2017-03	2022-06: Law	D, P
	Paraffin waxes and hydrocarbon waxes, chlorinated		63449-39-8	HAPS; 1907/2006; 2019/1021	ditto; Prohibited if short-chained chlorinated alkanes are present		2017-03	2022-06: Law	D, P
	Alkanes, C6-18, chlorinated		68920-70-7	HAPS; 1907/2006; 2019/1021	ditto; Prohibited if short-chained chlorinated alkanes are present		2017-03	2022-06: Law	D, P
	Alkanes, C12-13, chlorinated		71011-12-6	HAPS; 1907/2006; 2019/1021	ditto; Prohibited if short-chained chlorinated alkanes are present		2017-03	2022-06: Law	D, P
	Alkanes, C12-24, chlorinated		68527-02-6	HAPS; 1907/2006; 2019/1021	ditto; Prohibited if short-chained chlorinated alkanes are present		2019-09	2022-06: Law	D, P
	Alkanes, C14-17, chlorinated		85535-85-9	HAPS; 1907/2006	ditto; Prohibited for new developments $\geq 0,1$ % in mixture if medium-chain chloralalkanes are contained		2017-03	2022-06: Law, remark	D, P
	Alkanes, C18-20, chlorinated		106232-85-3	HAPS	ditto		2017-03		D
	Tetradecane, chloro derivs.		198840-65-2	HAPS; 1907/2006	ditto; Prohibited for new developments $\geq 0,1$ % in mixture if medium-chain chloralalkanes are contained		2017-03	2022-06: Compound, Law, Remark, Classification	D, P
	Alkanes, C22-30, chlorinated		288260-42-4	HAPS	ditto		2017-03		D

Name of group	Substance	Synonym	CAS number	Law/Regulation/Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Alkanes, C14-16, chlorinated		1372804-76-6	HAPS; 1907/2006	ditto; Prohibited for new developments $\geq 0,1$ % in mixture if medium-chain chloralalkanes are contained		2017-03	2022-06: Law, Remark, Classification	D, P
	Alkanes, C22-30, branched and linear, chlorinated		1401974-24-0	HAPS	ditto		2017-03		D
	Alkanes, C24-28, chlorinated		1402738-52-6	HAPS	ditto		2017-03		D
	Alkanes, C21-34, branched and linear, chlorinated		1417900-96-9	HAPS	ditto		2017-03		D
C10-C13 Chloroalkanes	short-chained chloroalkanes		85535-84-8	1907/2006	Compulsory declaration requirement for all applications not otherwise prohibited			2014-09: CAS number	D
	short-chained chloroalkanes		85535-84-8	1907/2006; 2019/1021	Prohibited in mixtures; discharge into sewage system prohibited	cleaning agents, solvents		2022-06: Remark	P
	2-Chloroethanol		107-07-3	1272/2008	substance absorbed by the skin with risk of teratogenic effect	Solvent		2018-08: Remark	D
	5-Chlor-2-methyl-2,3-dihydroisothiazol-3-one		55965-84-9	VKIS-ISI-IGM	In mixture 3:1 with 2-methyl-2,3-dihydroisothiazol-3-one labeling with H317 above 15 ppm. Risk of skin sensitization. Not suitable for pretreatment. Biocidal active substance	Cooling lubricant	2014-09	2016-03: Remark	D
Chlorofluorocarbons (CFCs/HCFCs)				1005/2009	Use prohibited due to potential for depletion of ozone layer	Aerosol propellant		2019-09: Group name, law	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Chlorinated Hydrocarbons (regulated)				1907/2006; ChemVerbotsV	Compulsory declaration requirement for all further aliphatic chlorinated hydrocarbons and, subject to specific restrictions, for use in industrial plants	cleaning agents, solvents		2014-09: Law	D
				1907/2006 Annex XVII No. 32-38; ChemVerbotsV; GB 24409 (CN)	Listed aliphatic chlorinated hydrocarbons not permitted as a substance and in mixtures for supply to the general public and/or if release may not be excluded. Ban on aliphatic chlorinated hydrocarbons > 0,1 % in paints	Cleaning agents, solvents, paints		2021-05: Law, remark, example of use	P
	e.g. Chloroform		67-66-3	1907/2006 Annex XVII No. 32; Clean Air Act (US)	ditto	cleaning agents, solvents		2022-06: Law	D, P
	1,2-Dichloroethane	Ethylene-chloride	107-06-2	1907/2006 Annex XIV, Annex XVII No. 28; Clean Air Act (US)	Prohibition without approval starting 22.11.2017	cleaning agents, solvents		2022-06: Law, remark	D, P
	1,1-Dichloroethane		75-34-3	1005/2009 Clean Air Act (US)	depletes ozone layer	cleaning agents, solvents		2022-06: Law; Remark	D, P
	1,1-Dichloroethylene		75-35-4	1907/2006 Annex XVII No. 38; Clean Air Act (US)	ditto	cleaning agents, solvents		2016-03: Remark	D, P
	Dichloromethane	Methylene-chloride	75-09-2	1907/2006 Annex XVII No. 59; Clean Air Act (US)	Prohibited above 0,1 % in paint strippers	cleaning agents, aerosols, color remover (paint remover, paint stripper), scouring (of metal and plastic forms), expanding agent (for foams).		2022-06: Law, remark	D, P
	Hexachloroethane		67-72-1	1907/2006 Annex XVII No. 41; Clean Air Act (US)	Prohibited for metal processing of non-iron metals			2014-09: Law, Classification	D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Pentachloroethane		76-01-7	1907/2006 Annex XVII No. 37	ditto	cleaning agents, solvents		2016-03: Remark	D, P
	Perchloroethylene	Tetrachloroethylene	127-18-4	1005/2009; Clean Air Act (US)	depletes ozone layer	cleaning agents, solvents		2022-06: Law, remark	D, P
	1,1,2,2-Tetrachloroethane		79-34-5	1907/2006 Annex XVII No. 35; Clean Air Act (US)	ditto	cleaning agents, solvents		2016-03: Remark	D, P
	1,1,1,2-Tetrachloroethane		630-20-6	1907/2006 Annex XVII No. 36	ditto	cleaning agents, solvents		2016-03: Remark	D, P
	Tetrachloromethane	Carbon-tetrachloride	56-23-5	1005/2009; ChemVerbotsV; Clean Air Act (US)	Prohibited in substances and mixtures; depletes ozone layer	cleaning agents, solvents		2022-06: Law, remark	D, P
	1,1,1-Trichloro-2-2-bis-(4-chlorophenyl)-ethane	DDT	50-29-3	ChemVerbotsV; 2019/1021	Prohibited in mixtures	cleaning agents, solvents		2019-09: Law, remark	D, P
	1,1,2-Trichloroethane		79-00-5	1907/2006 Annex XVII No. 34; Clean Air Act (US)	ditto	cleaning agents, solvents		2016-03: Remark	D, P
	1,1,1-Trichloroethane		71-55-6	1005/2009; TRGS 900; ChemVerbotsV; Clean Air Act (US)	Prohibited in substances and mixtures; depletes ozone layer, hazardous to water; substance absorbed by the skin with risk of teratogenic effect	cleaning agents, solvents		2022-06: Law	D, P
	Trichloroethylene	Trichloroethylene	79-01-6	1907/2006 Annex XIV, Annex XVII No. 28/29; Clean Air Act (US)	Prohibition without approval starting 21.04.2016	cleaning agents, solvents		2016-03: Law, Remark, Classification	P
	Vinyl chloride	Chloroethylene	75-01-4	1907/2006 Annex XVII No. 2; Clean Air Act (US)	Prohibited for aerosols			2014-09: Law	D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Chromium-VI-compounds				1907/2006; 2000/53/EG; Clean Air Act (US)	Compulsory declaration requirement for the exceptions as per 2000/53/EC Annex II	Corrosion protection layers		2014-09: Law	D
				1907/2006 Annex XVII No. 28/47; 2000/53/EG; GB 24409 (CN)	can cause cancer; Prohibited as a substance and in mixtures/paints; Prohibited in vehicles			2021-05: Law, remark	P
Cyclohexane			110-82-7	1907/2006	On certain conditions use by commercial customers permitted.	Solvent			D
			110-82-7	1907/2006 Annex XVII No. 57	Putting into circulation in neoprene-based contact adhesive is prohibited	Solvent		2014-09: Law	P
1,2-Cyclohexylene-dinitrilo tetraacetic acid	CDTA	13291-61-7			complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
Diethylene glycol		111-46-6	VKIS-ISI-IGM		Substance with limit value	Cooling lubricant	2014-09		D
Diethylenetriamine-pentaacetic acid	DTPA	67-43-6			complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
Diethylenetriamine-pentaacetic acid, Penta-sodium salt		140-01-2			complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
Diethylene-trinitilo penta-methylene phosphonic acid	DTPMP	15827-60-8			complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	1,6-Dihydroxy-2,5-dioxahexane		3586-55-8	VKIS-ISI-IGM	Biocidal active substance (formaldehyde depot substance) Residues of free formaldehyde in the biocidal product shall be indicated.	Cooling lubricant	2014-09		D
	N,N-Dimethylacetamide		127-19-5	1907/2006; TRGS 900	On certain conditions use by commercial customers permitted. Toxic for reproduction category 1B	Solvent		2018-08: Remark	D
			127-19-5	1907/2006 Annex XVII No. 30	Prohibited as a substance and in mixtures for supply to the general public; Toxic for reproduction category 1B	Solvent		2019-09: Remark	P
	N,N-Dimethylformamide		68-12-2	1907/2006; Clean Air Act (US)	On certain conditions use by commercial customers permitted.	Solvent		2014-09: Law	D
			68-12-2	1907/2006 Annex XVII No. 76	Prohibited as a substance and in mixtures as of 0,3 %; risk of impaired fertility	Solvent		2022-06: Law, remark	P
	3,5-Di-tert-butyl-4-hydroxytoluene	BHT	128-37-0	VKIS-ISI-IGM	Substance with limit value	Cooling lubricant	2014-09		D
	2-Ethoxyethanol	Ethyl glycol	110-80-5	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
		Ethyl glycol	110-80-5	1907/2006 Annex XVII No. 30; GB 24409 (CN)	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility; Prohibited > 0,03 % in paints (total content ethylene glycol ether/ester)	Solvent for paints		2021-05: Law, remark	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	2-Ethoxy-ethylacetate	Ethyl glycol-acetate	111-15-9	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
		Ethyl glycol-acetate	111-15-9	1907/2006 Annex XVII No. 30; GB 24409 (CN)	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility; Prohibited > 0,03 % in paints (total content ethylene glycol ether/ester)	Solvent for paints		2021-05: Law, remark	P
	Ethylbenzene		100-41-4	GB 24409 (CN)	Prohibited in China only > 1 % in water-based and > 30 % in solvent-based paints	Paints	2021-05	2022-06: Remark	P
	Ethylenediamine-tetraacetic acid	EDTA	60-00-4		Compulsory declaration requirement for applications outside the waste water system	Complexing agent			D
		EDTA	60-00-4		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals. Depending on the waste water system	Complexing agent		2022-06: Law, remark	P
	Ethylenediamine-tetraacetic acid, Tetra-sodium salt		64-02-8		Compulsory declaration requirement for applications outside the waste water system.				D
			64-02-8		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals. Depending on the waste water system	Complexing agent		2022-06: Law, remark	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Ethylenediamine-tetraacetic acid, Tri-sodium salt		7379-28-4		Compulsory declaration requirement for applications outside the waste water system	Complexing agent			D
			7379-28-4		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals. Depending on the waste water system	Complexing agent		2022-06: Law, remark	P
	Ethylenediamine-tetramethylene-phosphonic acid	EDTMP	1429-50-1		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	Ethylene glycol-bis(beta-aminoethylether)-N,N-tetraacetic acid	EGTA	67-42-5		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	Ethyl pyrrolidone	NEP	2687-91-4	1272/2008; 1907/2006 Annex XVII No. 30	Prohibited as a substance and in mixtures as of 0,3 %; toxic for reproduction category 1B; Use only in closed systems, except for purposes of research or analysis while not exceeding the specified required quantities.	Solvent for paints		2022-06: Remark	P
Fluorinated hydrocarbons				517/2014	Restrictions on usage due to Global Warming Potential, e.g. prohibited in foams and in technical aerosols with a GWP of more than or equal to 150	(Aerosol) propellants, refrigerants		2019-09: Group name, Law, Remark, Classification	D, P
	Hydrofluoric acid	Hydrogen fluoride	7664-39-3	Clean Air Act (US)	Sewage water exposure of heavy metals.	Additive in phosphating agents		2014-09: Law	D

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Formaldehyde		50-00-0	1907/2006; Clean Air Act (US)	Compulsory declaration requirement for all applications not otherwise prohibited; stringent requirements for emissions protection (refer to GS 93008-4)	Residues and degradation products of plastics (aminoplasts, urea- and melamine resins, foam plastics, vulcanization accelerators, basis for synthetic tannins, biocides (e.g. disinfectants), adhesives, pressed wood elements		2014-09: Law	
			50-00-0	ChemVerbotsV	Prohibited in detergents, cleaning agents and care products; danger of irreversible damage			2014-09: Law	P
Gluconates					complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	2,2',2''-(Hexahydro-1,3,5-triazine-1,3,5-triyl)triethanol		4719-04-4	VKIS-ISI-IGM	Mandatory labeling as H317 above 0,1 %. Biocidal active substance (formaldehyde depot substance). residues of free formaldehyde in the biocidal product shall be indicated.	Cooling lubricant	2014-09	2018-08: Remark	D
	1-Hydroxyethane-1,1-diphosphoric acid	HEDP	2809-21-4		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	Iso-Hexane	Iso-Hexane; 2-Methylpentane	107-83-5		Declarable due to residues of n-hexane in iso-hexane	Solvent	2016-03	2017-03: Remark, classification	D
	n-Hexane	n-Hexane	110-54-3	1907/2006 Annex XVII No. 30; Clean Air Act (US)	Prohibited as a substance and in mixtures > 5 %; risk of impaired fertility	Solvent		2019-09: Remark	D, P
	3-Iodo-2-propynyl butyl-carbamate	IPBC	55406-53-6	VKIS-ISI-IGM	Biocidal active substance	Cooling lubricant	2014-09	2016-03: Remark	D

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classific ation
Cobalt compounds, cobalt siccatives				1907/2006; Clean Air Act (US)	On certain conditions use by commercial customers permitted.	Pretreatment in paint shop, drying agents in paints		2014-09: Law	D
Cobalt compounds without cobalt siccatives				1907/2006 Annex XVII No. 28	Prohibited as a substance and in mixtures for supply to the general public; carcinogenic at inhalation; limit value 0,01 %			2019-09: Remark	P
e.g.	Cobalt dichloride		7646-79-9	1907/2006	ditto				D, P
	Cobalt sulfate		10124-43-3	1907/2006	ditto				D, P
	Cobalt acetate		71-48-7	1907/2006	ditto				D, P
	Cobalt nitrate		10141-05-6	1907/2006	ditto				D, P
	Cobalt carbonate		513-79-1	1907/2006	ditto				P
	2-Methoxy-ethanol	Methyl glycol	109-86-4	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
		Methyl glycol	109-86-4	1907/2006 Annex XVII No. 30; GB 24409 (CN)	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility; Prohibited > 0,03 % in paints (total content ethylene glycol ether/ester)	Solvent for paints		2021-05: Law, remark	P
	2-(2-Methoxyethoxy) ethanol	DEGME	111-77-3		On certain conditions use by commercial customers permitted.	Solvent for paints			D
		DEGME	111-77-3	1907/2006 Annex XVII No. 54	Putting into circulation in paints, paint removers, cleaning agents, self-shining emulsions or floor sealants is prohibited	Solvent for paints		2014-09: Law	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	2-Methoxy-ethylacetate	Methyl-glycol acetate	110-49-6	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
		Methyl-glycol acetate	110-49-6	1907/2006 Annex XVII No. 30; GB 24409 (CN)	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility; Prohibited > 0,03 % in paints (total content ethylene glycol ether/ester)	Solvent for paints		2021-05: Law, remark	P
	2-2-Methoxy-1-propanol		1589-47-5	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
			1589-47-5	1907/2006 Annex XVII No. 30	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility	Solvent for paints		2019-09: Remark	P
	2-Methoxy-1-propylacetate		70657-70-4	1907/2006	On certain conditions use by commercial customers permitted.	Solvent for paints			D
			70657-70-4	1907/2006 Annex XVII No. 30	Prohibited as a substance and in mixtures for supply to the general public; risk of impaired fertility	Solvent for paints		2019-09: Remark	P
	2-2-Methyl-2H-isothiazol-3-one		2682-20-4	VKIS-ISI-IGM	Biocidal active substance	Cooling lubricant	2014-09		D
	Methanol		67-56-1	1907/2006 Annex XVII No. 69	Prohibited in windshield washer fluid ≥ 0.6 % (w/w)	Windshield washer fluid	2016-03	2021-05: Law, remark	D, P
	N,N'-Methylene-bis-(5-methyloxazolidine)		66204-44-2	VKIS-ISI-IGM	Biocidal active substance (formaldehyde depot substance); residues of free formaldehyde in the biocidal product shall be indicated.	Cooling lubricant	2014-09		D

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Methyl glycine diacetic acid	MGDA	164462-16-2		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent	2016-03		D
	Monoethylamine	MEA	75-04-7		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent	2016-03		D
	Methyl pyrrolidone	NMP	872-50-4	1907/2006 Annex XVII No. 71	Prohibited as a substance and in mixtures as of 0,3 %; toxic for reproduction category 1B; Use only in closed systems, except for purposes of research or analysis while not exceeding the specified required quantities.	Solvent for paints		2022-06: Law, remark	P
	Methyl-t-butyl ether	MTBE	1634-04-4	Clean Air Act (US)	Air polluter			2018-08: Law, remark	D
	Monomethyl dibromo-diphenyl methane	DBBT	99688-47-8	1907/2006 Annex XVII No. 26	Prohibited as a substance and mixtures			2019-09: Remark	P
	Monomethyl dichloro-diphenyl methane	Ugilec 121 or Ugilec 21	81161-70-8	1907/2006 Annex XVII No. 25	Prohibited as a substance and mixtures			2019-09: Remark	P
	Monomethyl tetrachloro-diphenyl methane	Ugilec 141, Tetrachlor-benzyl-toluene	76253-60-6	1907/2006 Annex XVII No. 24	Prohibited as a substance and mixtures			2019-09: Remark	P
	Naphthalene		91-20-3	Clean Air Act (US)	Air polluter			2018-08: Law, remark	D
Naphthylamine, 2- and its salts				1907/2006	On certain conditions use by commercial customers permitted.				D
				1907/2006 Annex XVII No. 12/28	Prohibited as a substance and mixtures			2019-09: Remark	P
	e.g. 2-Naphthylamine		91-59-8	1907/2006	ditto				D, P
	N-Phenyl-2-naphthylamine		135-88-6	1907/2006	ditto				D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classific ation
Nickel				1907/2006 Annex XVII No. 27; Clean Air Act (US)	Prohibited in articles intended to come into direct and pro-longed contact with the skin, refer also to Table 2	Jewelry, flame spraying, special materials, constituent in metals		2014-09: Law, Remark, Usage, Classification	D, P
	e.g. Nickel alloys		7440-02-0	1907/2006	ditto			2017-03: Classification	D, P
	Nickel		7440-02-0	1907/2006	ditto			2017-03: Classification	D, P
Nickel compounds				1907/2006	On certain conditions use by commercial customers permitted. Carcinogenic for humans	Pretreatment in paint shop			D
				1907/2006 Annex XVII No. 28	Prohibited as a substance and in mixtures for supply to the general public			2019-09: Remark	P
e.g.	Dinickel trioxide		1314-06-3	1907/2006	ditto				D, P
	Nickel dioxide		12035-36-8	1907/2006	ditto				D, P
	Nickel monoxide		1313-99-1	1907/2006	ditto				D, P
	Nickel sulfate		16812-54-7	1907/2006	ditto				D, P
	Tetracarbonylnickel		13463-39-3	1907/2006	ditto				D, P
	Nickel sulfate		7786-81-4	1907/2006	ditto				D, P
	Nickel(II)carbonate		3333-67-3	1907/2006	ditto				D, P
	Carbonic acid, nickel salt		16337-84-1	1907/2006	ditto				D, P
	[μ-[Carbonato(2-)-O:O']] dihydroxytrinickel		65405-96-1	1907/2006	ditto				D, P
	[Carbonato(2-)] tetrahydroxytrinickel		12607-70-4	1907/2006	ditto				D, P
	Nickel dichloride		7718-54-9	1907/2006	ditto				D, P
	Nickel dinitrate		13138-45-9	1907/2006	ditto				D, P
	Nitric acid, nickel salt		14216-75-2	1907/2006	ditto				D, P
	Trinickel disulfide		12035-72-2	1907/2006	ditto				D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Nitrite				TRGS 615	Precursors for carcinogenic nitrosamines. Regulated according to TRGS 615 for corrosion inhibitors.	Corrosion protection agents, films			D
	e.g. Sodium nitrite		7632-00-0	TRGS 615	ditto	ditto			D
	Potassium nitrite		7758-09-0	TRGS 615	ditto	ditto			D
Nitrating agents				TRGS 615	Precursors for carcinogenic nitrosamines. Regulated according to TRGS 615 for corrosion inhibitors.	Corrosion protection agents, films			D
	e.g. Nitrous acid		7782-77-6	TRGS 615	ditto	ditto			D
	N-(2-Hydroxyethyl)-ethylene diamine-N,N,N-triacetic acid	HEDTA	150-39-0		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	Nitrilotriacetic acid	NTA	139-13-9		Compulsory declaration requirement for applications outside the waste water system				D
		NTA	139-13-9		Prohibited since the heavy metal load in the waste water treatment is increased.	Complexing agent			P
	Nitrilotriacetic acid-Trisodium salt		5064-31-3		Compulsory declaration requirement for applications outside the waste water system				D
			5064-31-3		Prohibited since the heavy metal load in the waste water treatment is increased.	Complexing agent			P
	4-Nitrodiphenyl		92-93-3	1907/2006; Clean Air Act (US)	On certain conditions use by commercial customers permitted.			2014-09: Law	D
			92-93-3	1907/2006 Annex XVII No. 14	Prohibited as a substance and mixtures			2019-09: Remark	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Nonylphenols			25154-52-3	1907/2006	Compulsory declaration requirement for all applications not otherwise prohibited	cleaning agents			D
			25154-52-3	1907/2006 Annex XVII No. 46a	Prohibited in cleaning agents, substances for textile, leather, paper and metal processing, formulating agent in pesticides, skin care products; discharge into sewage system prohibited	cleaning agents		2014-09: Law	P
e.g. 4-Nonylphenol			104-40-5 142731-63-3 17404-66-9 186825-36-5 26543-97-5 30784-30-6 52427-13-1 84852-15-3	1907/2006	ditto; Candidate list	ditto		2021-05: Classification	D, P
Pentachlorophenol and its salts				1907/2006 Annex XVII No. 22	Prohibited as a substance and in mixtures (limit value 0,1 %);	Wood preservatives, salts used to protect leathers and animal skins, stabilizers for latex		2019-09: Remark	P
e.g. Sodium-Pentachlorophenolate			131-52-2	ditto	ditto	ditto			P
Pentachlorophenol	PCP		87-86-5	ditto; Clean Air Act (US)	ditto	ditto		2014-09: Law, Classification	D, P
Perfluorooctanoic acid	PFOA			2019/1021	Prohibited as a substance and in mixtures (limit value 0,000 002 5 %) as an acid, salts and PFOA-related compounds	Fire extinguishing agent with specific exceptions, PTFE micropowder with specific limit value		2021-05: Law, remark, example of use, classification	D, P
Perfluorooctansulfonate	PFOS		1763-23-1 45298-90-6 and others	2019/1021	Prohibited as a substance and in mixtures (limit value 0,001 %) as an acid and its derivatives.	Galvanizing, in fire extinguishing agents and in coating of textiles		2021-05: CAS Number, Law, Remark, Classification	D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	2-Phosphonobutane-1,2,4-tricarboxylic acid	PBTC	37971-36-1		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
Phthalates				1907/2006 Annex XVII	On certain conditions use by commercial customers permitted.	Plasticizer		2022-06: Law	D
				1907/2006 Annex XIV	Prohibition without approval	Plasticizer		2018-08: Law, remark	P
	Diisodecyl phthalate	DIDP	26761-40-0 68515-49-1	1907/2006 Annex XVII No. 52	Prohibited in toys and articles that may be swallowed by children	ditto		2022-06: Law	D, P
	Diisononyl phthalate	DINP	28553-12-0 68515-48-0	1907/2006 Annex XVII No. 52	Prohibited in toys and articles that may be swallowed by children	ditto		2022-06: Law	D, P
	Di-n-octylphthalate	DOP	117-84-0	1907/2006 Annex XVII No. 52	Prohibited in toys and articles that may be swallowed by children	ditto		2018-08: Law, remark	D, P
Polybrominated diphenyl ether				1907/2006	Compulsory declaration requirement for all substances not otherwise prohibited	Flame retardant in plastic parts and textiles			D
				1907/2006 Annex XVII No. 45	Ban on octa-brominated compounds	Flame retardant in plastic parts and textiles		2014-09: Law	P
	e.g. Decabromodiphenyl ether	Bis-(pentabromo phenyl) ether	1163-19-5	1907/2006 Annex XVII No. 67	Prohibited as a substance and in mixtures as from 02 March 2019, exceptions apply for particular spare parts	ditto		2021-05: Classification	D, P
	Octabromodiphenyl ether	Diphenyl ether octabromo-derivative	32536-52-0	1907/2006	Prohibited in substances and mixtures; discharge into sewage system prohibited	ditto		2019-09: Remark	P
	Pentabromodiphenyl ether	Diphenyl-ether pentabromo derivative	32534-81-9	2019/1021	Prohibited as substance and in mixtures (analog also tetra-, hexa- and hepta bromine derivatives)	ditto		2019-09: Law, remark	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
	Polytetrafluoroethylene	Teflon PTFE	9002-84-0		The use of Teflon is prohibited in all paint shops, upstream areas (press shop and add-on parts, body shop) or neighboring areas (social and sanitary areas) as well as in the range of the bonding stations, e.g. high voltage batteries, e-machine and power electronics.	Seals, plain bearings		2022-06: Synonym, Remark	D, P
	Propylene-diamine tetraacetic acid		1939-36-2		Compulsory declaration requirement for applications outside the waste water system			2022-06: Law	D
			1939-36-2		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent		2022-06: Law, remark	P
	1,3-Propane sultone		1120-71-4	GefStoffV	Use only in closed systems, except for purposes of research or analysis while not exceeding the specified required quantities.	Additive in LIB	2019-09	2021-05: Classification	D, P
Mercury and its compounds				1907/2006; 2000/53/EG; Clean Air Act (US)	Compulsory declaration requirement for the exceptions as per 2000/53/EC Annex II	Gas discharge lamps, display illumination		2014-09: Law	D
				1907/2006 Annex XVII No. 18; 2000/53/EG; GB 24409 (CN)	Prohibited in vehicles, prohibited in water treatment, prohibited in paints			2021-05: Law, remark	P
	e.g. Mercury		7439-97-6	1907/2006; 2000/53/EG	ditto				D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Hydrogen sulfide emitting substances				Odor in vehicle parts	Breakdown products in vulcanization				D
	e.g. Hydrogen sulfide	Hydrogen sulfide	7783-06-4	Clean Air Act (US)	ditto	ditto		2014-09: Law	D
	Tensides not fully biodegradable			648/2004	Tensides which are not fully biodegradable according to the detergent regulation 648/2004 are prohibited in detergents.	cleaning agents			P
	Toluene		108-88-3	1907/2006; TRGS 900; Clean Air Act (US)	On certain conditions use by commercial customers permitted. Toxic for reproduction category 2			2018-08: Remark	D
			108-88-3	1907/2006 Annex XVII No. 48; GB 24409 (CN)	Prohibited in adhesives $\geq 0,1$ % and spray paints for supply to the general public Prohibited > 1 % in paints			2021-05: Law, remark	P
	Trichlorobenzene		12002-48-1	1907/2006	On certain conditions use by commercial customers permitted.				D
			12002-48-1	1907/2006 Annex XVII No. 49	Prohibited as a substance and in mixtures			2019-09: Remark	P
	Triethanolamine	2,2',2''-Nitrilo-triethanol	102-71-6		complexing agent, biologically hardly degradable. Sewage water exposure of heavy metals.	Complexing agent			D
	1,3,5-Tris-(2-hydroxypropyl)-hexahydro-1,3,5-triazine		25254-50-6	VKIS-ISI-IGM	Biocidal active substance; Residues of free formaldehyde in the biocidal product shall be indicated	Cooling lubricant	2014-09		D
	Xylene		1330-20-7	GB 24409 (CN)	Prohibited in China only > 1 % in water-based and > 30 % in solvent-based paints	Paints	2021-05	2022-06: Remark	P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Tin organic compounds				1907/2006; TRGS 900	Compulsory declaration requirement for all applications not otherwise prohibited	Paints		2019-09: Remark	D, P
				1907/2006 Annex XVII No. 20	Prohibition of all organostannic substances in protective paint coats for maritime applications, in water treatment and at discharge into sewage. Tri-substituted organostannic substances is prohibited in articles when concentration of tin is > 0,1 %. Dibutyltin (DBT) substances are prohibited in mixtures and articles for putting into public circulation where the concentration of tin is > 0,1 %. Tin compounds containing tin > 0,1 % are prohibited in certain articles (e.g. textiles, gloves, two-component room temperature vulcanization molding kits).				
	e.g. Di-μ-oxo-di-n butyl-stanniohydroxyborane	DBB	75113-37-0	ditto	ditto	ditto			D, P
	Dibutyltin-bis-(maleic acid isobutyl ester)	Dibutyl-tin-dilaurate	77-58-7	ditto	ditto	ditto			D, P
	Dibutyl tin dichloride		683-18-1	ditto	ditto	ditto			D, P
	Dibutyl tin oxide		818-08-6	ditto	ditto	ditto			D, P

Name of group	Substance	Synonym	CAS number	Law/ Regulation/ Guideline	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classific ation
	Tributyl tin benzoate		4342-36-3	ditto	ditto	ditto			D, P
	Tributyl tin chloride		1461-22-9	ditto	ditto	ditto			D, P
	Tributyl tin fluoride		1983-10-4	ditto	ditto	ditto			D, P
	Tributyl tin methacrylate		2155-70-6	ditto	ditto	ditto			D, P
	Tributyl tin naphthenate		85409-17-2	ditto	ditto	ditto			D, P
	Tributyl tin oleate		24124-25-2	ditto	ditto	ditto			D, P
	Tributyl tin oxide	TBTO	56-35-9	ditto	ditto	ditto			D, P
	Tributyl tin compounds		688-73-3	ditto	ditto	ditto			D, P

Table 1 b: Regulated substance groups and product groups

Group name	Law/ regulation/ guideline/ study	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Skin sensitizing products and products with caustic potential to skin	1272/2008	Compositions classified according to 1272/2008 with H317, or H314 and for which skin contact or absorption via the respiratory tracts may not be ruled out. The classification as H317, or H314 shall relate to the overall product, the evaluation and hence the possible triggering of a search for a substitute substance shall therefore not relate to a single substance.	cleaning agents		2022-06: Group name, remark	D
Silicone-containing products		The use of silicone materials is prohibited in all paint shops, upstream areas (press shop and add-on parts, body shop) or neighboring areas (social and sanitary areas) as well as in the range of the bonding stations, e.g. high voltage batteries, e-machine and power electronics. This restriction applies in plant Leipzig for all areas. This product group includes silicones and siloxanes but not silanes and silicates.	Defoaming agent, glass cleaning agents		2022-06: Remark	D, P
Compounds classified as C, M, R class 2 (according to 1272/2008) and/or mixtures with their overall classification	1272/2008				2019-09: Group name	D
Compounds classified as C, M, R class 1A or 1B (according to 1272/2008) or class 1 or 2 according to TRGS 905 and/or mixtures with their overall classification	1907/2006 Annex XVII No. 28/29/30	Prohibited as a substance and in mixtures for supply to the general public Use by commercial customers permitted under certain conditions; prohibited for new developments depending on the concentration.			2021-05: Classification	D, P
Compounds according to 1907/2006 on the Annex XIV	1907/2006 Annex XIV	Substances on Annex XIV are prohibited after expiry of the sunset date without authorization.		2016-03	2018-08: Group name, law	P

Group name	Law/ regulation/ guideline/ study	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classification
Compounds classified according to TRGS 900 as H, Y	TRGS 900	Skin-resorptive substance that does not pose a risk of damaging the fetus if the occupational exposure limit value (OELV) and biological limit value (BLV) are complied with. The hazard analysis of the workplace results in measures for compliance with the air pollution limit values and for skin protection; in these cases, it is not necessary to initiate an additional search for substitute substances based on the properties of the substances at the workplace.			2019-09: Remark	D
Substances classified according to TRGS 900 as H, Z	TRGS 900	Skin-resorptive substance in which a risk of fetal damage may not be excluded, even if the OELV and BLV are complied with. H, Z substances are already listed in other categories or as a single substance due to their toxic effect on reproduction		2014-09	2022-06: Remark, Example of use	D
Substances marked in accordance with 1272/2008 with the GHS06 symbol (skull and crossbones) (H300, H301, H310, H311, H330, H331)	1272/2008				2018-08: Group name	D
Compounds classified as water endangering class WGK 3	AwSV	Severely hazardous to water. The Water Hazard Classifications relate to the overall product, the evaluation and hence the possible trigger of a search for a substitute substance shall therefore not relate to a single substance.			2022-06: Law	D
Substances with STOT SE/RE Cat. 1, 2 (H370, H371, H372, H373)	1272/2008	Target organ toxicity		2016-03	2018-08: Group name	D
Per and polyfluoralkyl substances	1907/2006	Declarable to limit of detection	Leather and textile cleaning products	2022-06		D
Products containing diisocyanate	1907/2006 Annex XVII No. 74	Obligatory training and certification of all users in Europe for safe handling of products containing diisocyanate $\geq 0,1\%$; First training to be completed until August 24, 2023 and then every 5 years	Polyurethane adhesives and sealants	2022-06		D

Table 2: Substances and substance groups subject to substitution

Group name	Law/ regulation/ guideline/ study	Remark and regulated usage	Examples for usage	Inclusion in GS	Last amendment	Classifica tion
Compounds classified according to 1907/2006 as substances on the candidate list (Substances of Very High Concern, SVHC) $\geq$ 0,1 % in mixtures	1907/2006	prohibited for new developments		2021-05		S
Substances as such or in mixtures that result in an overall classification of the product with the following H statements: H300, H301, H310, H311, H330, H331, H334, H340, H341, H350s, H351, H360s, H361s, H370 and/or H372	TRGS 600; TRGS 905; 1272/2008			2021-05		S

A (Annex normative)

The exact chemical composition to ensure industrial hygiene purposes shall be requested using the letter on the following page.

The supplier shall complete the form "Exact chemical composition for industrial hygiene purposes" on page 41 and return it to the sender.

BMW AG, 80788 München

Your/our reference

Your message from

Department/ From

Telephone

Fax

e-mail

Date

Subject

TP-343
+49-89-382-51553

Brigitte.Stein-Schaller@bmw.de

Chemical product release

Dear Sir / Madam,

To ensure industrial hygiene purposes, we need the exact chemical composition of the products procured by you that are listed in the annex.

We assure you that your information will be treated with the strictest confidentiality and used for this purpose only.

To guarantee confidentiality please send the enclosed form only to:

Dr. Brigitte Stein-Schaller, Department TP-343, Telephone +49 89 382-51553

She is authorized to receive and manage the strictly confidential data and can be contacted for any further information in this context.

We thank you for returning the form as soon as possible

Yours sincerely

Bayerische Motoren Werke Aktiengesellschaft

Company

Bayerische Motoren Werke Aktiengesellschaft

Postal address BMW AG

80788 München

Office address Petuelring 130

Office address Forschungs- und
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**Chairman of the
Supervisory Board**

Norbert Reithofer

Board of Management

Oliver Zipse

(Chairman)

Ilka Horstmeier

Milan Nedeljković

Pieter Nota

Nicolas Peter

Joachim Post

Frank Weber

**Exact chemical composition
for industrial hygiene purposes**

Confidential

Return to:

Your address:

Mrs.
Dr. Brigitte Stein-Schaller
BMW Group
TP-343

D-80788 Munich

Name _____

Company _____

Department _____

Address _____

Telephone _____

Fax _____

Trade name: _____

Supplier material number (recipe number): _____

CAS number	% Weight	% Volume	Chemical identification

Date

Signature